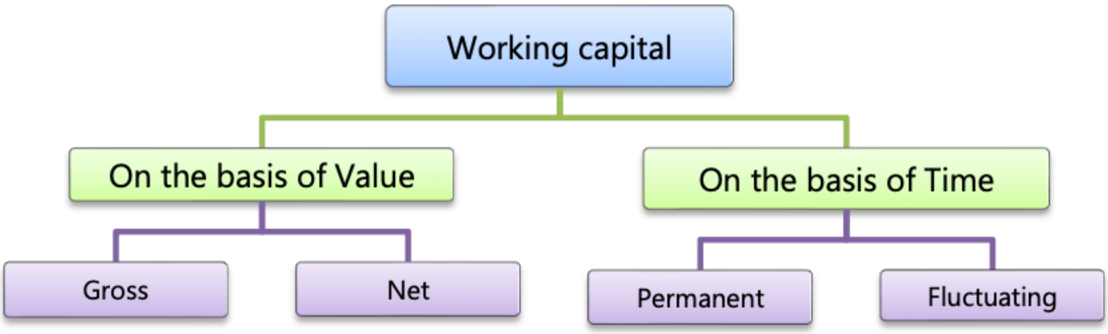
WORKING CAPITAL MANAGEMENT

1. Working Capital

Working capital is defined as the difference between current assets and current liabilities.



Gross Working Capital – It refers to the firm's investment in current assets.

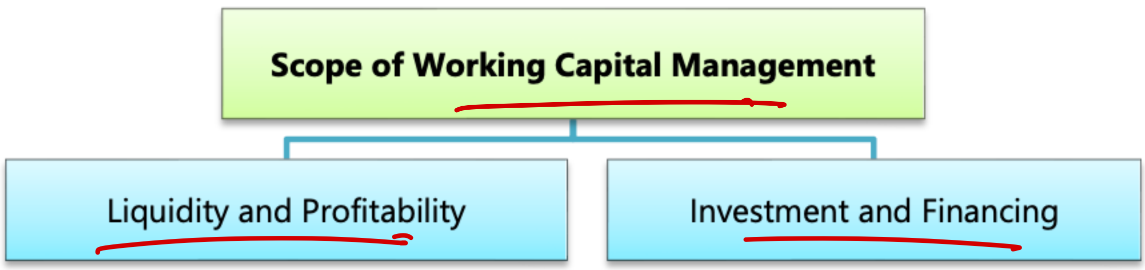
Net Working Capital – It refers to the difference between current assets and current liabilities.

Permanent Working Capital – It is the base working capital, which is the minimum level of investment in the current assets that is carried by the entity at all times to carry its day-to-day activities.

Temporary Working Capital – It is the amount which is used to finance the short-term working capital requirements which arises due to fluctuations in sales volume. It is also known as variable or fluctuating working capital.

2. Working Capital Management

It is a process which is designed to ensure that an organization operates efficiently by monitoring & utilizing its current assets and current liabilities to the best effect.



3. Estimation of Working Capital

(A) Component wise i.e. by preparing statement of WC estimation

(B) Operating cycle method

4. Statement of WC Estimation

Particulars	Amount
Raw material Stock $\left[\frac{Annual\ RM\ Consumption}{365\ /\ 52/\ 12} \times RM\ Stock\ period\right]$	
WIP Stock $\left[\left\{\frac{Annual\ RM\ Consumption}{365/52/12} \times WIP\ period \times DOC\right\} + \left\{\frac{Conversion\ cost}{365\ /\ 52/\ 12} \times WIP\ period \times DOC\right\}\right]$	
Finished goods stock $\left[\frac{COP}{365\ /\ 52/\ 12} \times FG\ Stock\ period\right]$	
Debtors $\left[\frac{Credit\ COS}{365\ /\ 52/\ 12} \times Average\ Collection\ period\right]$	
Prepaid expenses $\left[\frac{Expenses}{365\ /\ 52/\ 12} \times Prepaid\ Expense\ period\right]$	
Cash & Bank balance	
Total Current Assets (A)	
Creditors for Material $\left[\frac{Annual\ RM\ Purchase}{365\ /\ 52/\ 12} \times Creditors\ Payment\ period\right]$	
Outstanding expenses $\left[\frac{Expenses}{365\ /\ 52/\ 12} \times Outstanding\ expense\ period\right]$	
Total Current Liabilities (B)	
Working Capital (A – B)	
Add: Safety Margin	
Net Working Capital	

5. Points to Remember (PTRs)

- If Degree of Completion (DOC) of WIP is not given then assume to be at

  100% for Material

  50% for labour

  50% for direct expenses

  and 50% for factory/production overheads

Conv. Cost → 50%
Mat. → 100%

- Conversion cost = Direct labour + Direct expenses + Factory overheads

- Debtors can be computed on the basis of total credit sales although preferred to computed on basis of total credit cost of sales.

6. Cash Cost of WC

- Do not consider non-cash expenses e.g. depreciation while preparing income statement.

- Do not consider non-cash expenses for calculation of items of WC.

Question – 1	

COGS = 27L − 20% = 21.6L (Inc. Dep.)

SK Ltd. sells goods at a gross profit of 20%. It includes depreciation as part of cost of production. The following figures for the 12 months period ending 31st Dec. 2020 are given to enable you to ascertain the requirements of working capital of the company on a cash cost basis. In your workings, you are required to assume that:

(a) a safety margin of 15% will be maintained

(b) cash is to be held to the extent of 50% of current liabilities

(c) there will be no work-in-progress

(d) tax is to be ignored

Stocks of raw materials and finished goods are kept at one month's requirements

All working notes are to form part of your answer.

	₹
Sales – at 2 months credit	27,00,000
Materials consumed (suppliers credit is for 2 months)	6,75,000
Wages (paid at the beginning of the next month)	5,40,000
Manufacturing expenses outstanding at the end of the year (cash expenses are paid one month in arrear)	60,000
Administrative expenses (paid as above)	1,80,000
Sales promotion expenses – paid quarterly and in advance	90,000

Mat.	6,75,000
Cash Wages	5,40,000
(60000×12) Manuf. Exp.	7,20,000
COP/COGS	19,35,000
(+) Adm.	1,80,000
(+) S&D	90,000
COS	22,05,000
RM St. [6.75L × 1/12]	56250
WIP St.	—
FG St. [19.35L × 1/12]	161250
Drs. [22.05L × 2/12]	367500
Sell. Ad. [90000 × 1/4]	22500
Cash [50% × 232500]	116250
CA	723750
Crs. Mat. [6.75L × 2/12]	112500
O/S Wages [5.40L × 1/12]	45000
O/S M. Exp.	60000
O/S A. Exp. (1.80L × 1/12)	15000
CL	232500
WC (A−B)	491250
(+) SM @ 15%	
WC	564938

Solution

Working note:

Statement of cost of sales

Particulars	Amount
Raw material	6,75,000
Wages	5,40,000
Manufacturing expenses (60,000 × 12)	7,20,000
COP/COGS	19,35,000
Administrative expenses	1,80,000
Sales promotion expenses	90,000
Cost of sales (COS)	22,05,000

Statement showing Working Capital Requirements

Current Assets in terms of Cash Cost	Amount (₹)
Stock of Raw Material (6,75,000 x 1/12)	56,250
Stock of Finished Goods (19,35,000 × 1/12)	1,61,250
Debtors (22,05,000 × 2/12)	3,67,500
Prepaid sales promotion expenses (90,000 × 3/12)	22,500
Cash at Bank and in hand (50% × 2,32,500)	1,16,250
Total Current Assets (A)	**7,23,750**
Current Liabilities in terms of Cash cost	
Creditors for Material (6,75,000 × 2/12)	1,12,500
Creditors for Wages (5,40,000 × 1/12)	45,000
Creditors for Manufacturing Expenses (7,20,000 × 1/12)	60,000
Creditors for Administrative Expenses (1,80,000 × 1/12)	15,000
Total Current Liabilities (B)	**2,32,500**
Net Current Assets (A - B)	4,91,250
Add: 15% margin for contingencies	73,688
Required working capital	**5,64,938**

X, Y } → 100%
Z → ½ of Z → DOC 100%
Z → ½ of Z → DOC 50%

Question – 2	

While applying for financing of working capital requirements to a commercial bank, SK Ltd. projected the following information for the next year.

Cost Element	Per unit (₹)	Per unit (₹)	
Raw Materials			
X	30		45 L
Y	7		10.50 L
Z	6	43	9 L
Direct Labour		25	37.50 L
Manufacturing and administration overheads (excluding depreciation)		20	30 L
		COP/COGS	132 L
Depreciation		10	×
Selling overheads		15	22.50 L
		113	154.50 L

Additional Information:

(a) Raw materials are purchased from different suppliers leading to different period allowed as follows:
X – 2 months; Y – 1 months; Z – ½ month Crs.

(b) WIP Production cycle is of ½ month. Production process requires full unit of X and Y in the beginning of the production. Z is required only to the extent of half unit in the beginning and the remaining half unit is needed at a uniform rate during the production process.

(c) X is required to be stored for 2 months and other materials for 1 month.

(d) Finished goods are held for 1 month.

(e) 25% of the total sales is on cash basis and remaining on credit basis. The credit allowed by debtors is 2 months.

(f) Average time lag in payment of all overheads is 1 months and ½ months for direct labour.

(g) Minimum cash balance of ₹ 8,00,000 is to be maintained.

Calculate the estimated working capital required by the company on cash cost basis if the budgeted level of activity is 1,50,000 units for the next year. The company also intends to increase the estimated working capital requirement by 10% to meet the contingencies. (You may assume that production carried on evenly throughout the year and direct labour and other overheads accrue similarly.)

Solution

Statement showing Working Capital Requirements of

Current Assets	**Amount (₹)**
Stock of raw material X (45,00,000 × 2/12)	7,50,000
Stock of raw material Y (10,50,000 × 1/12)	87,500
Stock of raw material Z (9,00,000 × 1/12)	75,000
Stock of work-in-progress (working note – 2)	4,00,000
Stock of finished goods (1,32,00,000 × 1/12)	11,00,000
Debtors for credit sale (1,54,50,000 × 75% × 2/12)	19,31,250
Cash	8,00,000
Total Current Assets (A)	**51,43,750**

Current Liabilities	
Creditors for raw material X (45,00,000 × 2/12)	7,50,000
Creditors for raw material Y (10,50,000 × 1/12)	87,500
Creditors for raw material Z (9,00,000 × 0.5/12)	37,500
Outstanding direct labour (37,50,000 × 0.5/12)	1,56,250
Outstanding manufacturing & administration overheads (30,00,000 × 1/12)	2,50,000
Outstanding selling overheads (22,50,000 × 1/12)	1,87,500
Total Current Liabilities (B)	**14,68,750**
Net working capital (A – B)	**36,75,000**
Add: Provision for Contingencies @ 10%	3,67,500
Working capital requirement	**40,42,500**

Working Note-1

Statement of Cost

Particulars	₹
Raw material X (1,50,000 × 30)	45,00,000
Raw material Y (1,50,000 × 7)	10,50,000
Raw material Z (1,50,000 × 6)	9,00,000
Raw material consumed	64,50,000
Add: Direct labour (1,50,000 × 25)	37,50,000
Add: Manufacturing & Administration overheads (1,50,000 × 20)	30,00,000
Cash GFC/NFC/COP/COGS	1,32,00,000
Add: Selling overheads (1,50,000 × 15)	22,50,000
Cash cost of sales	1,54,50,000

Working Note-2

Statement of calculation of WIP

Particulars	₹
Raw material X (45,00,000 × 0.5/12)	1,87,500
Raw material Y (10,50,000 × 0.5/12)	43,750
Raw material Z (9,00,000 × 50% × 0.5/12)	18,750
Raw material usage	2,50,000
Add: Raw material Z (9,00,000 × 50% × 50% × 0.5/12)	9,375
Add: Direct labour (37,50,000 × 50% × 0.5/12)	78,125
Add: Manufacturing & Administration overheads (30,00,000 × 50% × 0.5/12)	62,500
Work in progress stock	4,00,000

Question – 3	

The management of SK Ltd. is planning to expand its business and consults you to prepare an estimated working capital statement. The records of the company reveal the following annual information:

	(₹)
Sales – Domestic at one month's credit	18,00,000
Export at three month's credit (sale price 10% below domestic price)	8,10,000
Materials used (suppliers extend two months credit)	6,75,000
Lag in payment of wages – ½ month	5,40,000
Lag in payment of manufacturing expenses (cash) – 1 month	7,65,000
Lag in payment of administration expenses – 1 month [Rel. to Prod.]	1,80,000
Selling expenses payable quarterly in advance	1,12,500
Income tax payable in four instalments of which one falls in the next financial year	1,68,000

19.80L

21.60L

Rate of gross profit is 20%. Ignore work-in-progress and depreciation. The company keeps one month's stock of raw materials and finished goods (each) and believes in keeping ₹ 2,50,000 available to it including the overdraft limit of ₹ 75,000 not yet utilized by the company.

Cash = 2.50L – 0.75L = 1.75L

The management is also of the opinion to make 10% margin for contingencies on computed figures. You are required to prepare the estimated working capital statement for the next year.

Solution

Statement of Working Capital Estimation

Particulars	Amount
Raw material stock (6,75,000 × 1/12)	56,250
Finished goods stock (21,60,000 × 1/12)	1,80,000
Domestic sales debtors (15,17,586 × 1/12)	1,26,466
Export sales debtors (7,54,914 × 3/12)	1,88,729
Prepaid selling expenses (1,12,500 × 3/12)	28,125
Cash and bank [2.50L – 0.75L]	1,75,000
Total CA (A)	7,54,570
Creditors for material (6,75,000 × 2/12)	1,12,500
Outstanding wages (5,40,000 × 0.5/12)	22,500
Outstanding manufacturing exp. (7,65,000 × 1/12)	63,750
Outstanding admin. Exp. (1,80,000 × 1/12)	15,000
Outstanding income tax (1,68,000 × ¼)	42,000
Total CL (B)	2,55,750
Working capital requirement (A – B)	4,98,820
Add: Margin of safety @10%	10% 49,882
Working capital requirement after margin	5,48,702

Working Note – 1:

Let domestic selling price = ₹ 100

Thus, Gross Profit on Domestic Sale = 100 × 20% = ₹ 20

Cost of Goods Sold on Domestic Sale = 100 – 20 = ₹ 80

Also export selling price = 100 – 10% = ₹ 90

Gross profit on export sales = 90 – 80 = ₹ 10

Working Note – 2:

Particulars	Domestic	Export	Total
Sales	18,00,000	8,10,000	26,10,000
(-) Gross Profit	18l × 20% = 3,60,000	8.10L × 10/90 = 90,000	4,50,000
COGS	14,40,000	7,20,000	21,60,000
(+) Selling Exp. (1,12,500 in 18:8.10)	77,586	34,914	1,12,500
Cost of Sales	15,17,586	7,54,914	22,72,500

7. Existing Business Vs New Business

Existing Business
- We assume that opening and closing stock will be same
- Don't show stock in income statement

New Business
- In this case, Opening stock = Nil but closing stock will be there
- Show closing stock in income statement

Question – 4

SK Limited is commencing a new project for manufacture of a plastic component. The following cost information has been ascertained for annual production of 12,000 units which is the full capacity:

	Cost per unit (₹)
Materials	40
Direct labour and variable expenses	20
Fixed manufacturing expenses	6
Depreciation	10
Fixed administration expenses (rel. to Prod.)	4
	80

The selling price per unit is expected to be ₹ 96 and the selling expenses ₹ 5 per unit, 80% of which is variable. In the first two years of operations, production and sales are expected to be as follows:

V F

Year	Op. St	Production (no. of units)	Sales (no. of units)	Cl. St
1	0 +	6,000 −	5,000 =	1000
2	1000 +	9,000 −	8,500 =	1500

To assess the working capital requirements, the following additional information is available:

(a) Stock of materials — 2.25 months average consumption
(b) Work-in-progress — Nil
(c) Debtors — 1 months average sales
(d) Cash balance — ₹ 10,000
(e) Creditors for supply of materials — 1 months average purchase during the year
(f) Creditors for expenses — 1 months average of all expenses during the year

Prepare, for the two years:

(i) A Projected statement of Profit/Loss (Ignoring taxation)
(ii) A Projected statement of working capital requirements

Solution

(i) Projected Statement of Profit & Losses

Particulars	Year 1	Year 2
Opening Stock of Raw Material	-	45,000
Add: Purchases (Bal. fig.)	2,82,000	3,82,600
(-) Closing Stock of Raw Material	(45,000)	(67,500)
Year 1 = (2,40,000 × 2.25/12)		
Year 2 = (3,60,000 × 2.25/12)		
Raw Material Consumed	2,40,000	3,60,000
(+) Direct Labour & Variable Expenses @ ₹ 20	1,20,000	1,80,000
Prime Cost	3,60,000	5,40,000

(+) Fixed Manufacturing (12,000 × 6)	72,000	72,000
(+) Depreciation (12,000 × 10)	1,20,000	1,20,000
(+)Fixed Administration expenses (12,000 × 4)	48,000	48,000
Cost of Production	6,00,000	7,80,000
(+) Opening Stock of Finished Goods	-	1,00,000
(-) Closing Stock of Finished Goods Year 1 = (6,00,000/6,000 × 1,000) Year 2 = (7,80,000/9,000 × 1,500)	(1,00,000)	(1,30,000)
Cost of Goods Sold	5,00,000	7,50,000
(+) Selling & Distribution overhead		
- Variable (5,000 × 4) (8,500 × 4)	20,000	34,000
- Fixed (12,000 × 1)	12,000	12,000
	32000	46000
Cost of Sales	5,32,000	7,96,000
(+) Profit/(loss) (Bal. fig.)	(52,000)	20,000
(5,000 × 96) (8,500 × 96) **Sales**	4,80,000	8,16,000

Projected Statement of Working Capital Requirement

Particulars	Year 1	Year 2
Current Assets		
Inventory of Raw Material	45,000	67,500
Inventory of finished goods	1,00,000	1,30,000
Debtors (5,32,000 × 1/12) (7,96,000 × 1/12)	44,333	66,333
Cash	10,000	10,000
Total (A)	1,99,333	2,73,833
Current Liabilities: Creditors for Material (Purch. Basis) (2,85,000 × 1/12) (3,82,500 × 1/12)	23,750	31,875
Creditors for expenses 120000 + 72000 + 48000 + 32000 Year 1: {~~4,80,000 + 2,88,000 + 1,92,000 + 1,28,000~~} × 1/12	22,667	
Year 2: {~~7,20,000 + 2,88,000 + 1,92,000 + 1,84,000~~} × 1/12 180000 + 72000 + 48000 46000		28,833
Total (B)	46,417	60,708
Net working Capital (A) – (B)	1,52,916	2,13,125

Question – 5

SK Ltd a company newly commencing business in 2021 has the under mentioned projected Profit & Loss Statement:

	₹	₹
Sales		2,10,000
Cost of goods sold		(1,53,000)
Gross profit		57,000
Administrative Expenses (Gen.)	14,000	
Selling Expenses	13,000	27,000
Profit before tax		30,000
Provision for taxation		(10,000)
Profit after tax		20,000
The cost of goods sold has been arrived at as under:		
Material used	84,000	
Wages & Manufacturing expenses	62,500	
Depreciation	23,500	
	1,70,000	
Less: Stock of finished goods (10% of goods produced not yet sold)	(17,000)	
	1,53,000	

The figures given above relate only to finished goods not to work-in-progress. Goods equal to 15% of the year's production (in terms of physical units) will be in process on the average requiring full materials but only 40% of the other expenses. The company believes in keeping materials equal to two months consumption in stock.

All expenses will be paid one month in advance. Suppliers of materials will be extending 1 – ½ months credit. Sale will be 20% for cash and the rest at two month's credit. 70% of the income tax will be paid in advance in quarterly instalments. The company wished to keep Rs 8,000 in cash. Prepare an estimate of working capital on cash cost basis.

Solution

Working Note: Projected Statement of Profit & Losses

Particulars	**Cash Cost Basis**
Opening stock of material	-
(+) Purchases (B/F)	1,12,700
(-) Closing stock of material (96,600 × 2/12)	(16,100)
Material Consumed [84,000 + (84,000 × 15%)]	96,600
Wages & Manuf. Exp. [62,500 + (62,500 × 15% × 40%)]	66,250
Depreciation [23,500 + (23,500 × 15% × 40%)]	-

Gross Factory Cost	1,62,850
(+) Opening WIP	-
(-) Closing WIP	(16,350)
NFC/COP	1,46,500
(+) Opening stock of FG	-
(-) Closing stock of FG	(14,650)
COGS	1,31,850
(+) Administrative expenses	14,000
(+) Selling expenses	13,000
Total Cost	1,58,850

Statement of Working Capital Estimation

Particulars	Cash Cost Basis
Current Assets	
Inventory of Raw Material	16,100
Inventory of WIP	16,350
Inventory of finished goods	14,650
Debtors (1,58,850 × 80% × 2/12)	21,180
Prepaid expenses [(66,250 + 14,000 + 13,000) × 1/12]	7,771
Advance tax paid (10,000 × 70% × 3/12)	1,750
Cash	8,000
Total (A)	85,801
Current Liabilities:	
Creditors for Material (1,12,700 × 1.5/12)	14,087
Outstanding income tax (10,000 × 30%)	3,000
Total (B)	17,087
Net working Capital (A) – (B)	68,714

Question – 6

Day Ltd. a newly formed company has applied to the Private Bank for the first time for financing its working capital requirements. The following information are available about the projects for the current year:

Estimated level of activity	Completed Units of Production 31,200 plus unit of work in progress 12,000
Raw Material Cost	₹ 40 per unit
Direct Wages Cost	₹ 15 per unit
Overheads	₹ 40 per unit (inclusive of depreciation ₹ 10 per unit)
Selling price	₹ 130 per unit

Raw material in stock	Average 30 days consumption
Work in Progress stock	Material 100% and Conversion cost 50%
Finished goods stock	24,000 units
Credit allowed by the supplier	30 days
Credit allowed to purchases	60 days
Direct wages (lag in payment)	15 days
Expected cash balance	₹ 2,00,000

Assume that production is carried on evenly throughout the year (360 days) and wages and overheads accrue similarly. All sales are on the credit basis. You are required to calculate the Net Working Capital Requirement on Cash Cost Basis.

Solution

Statement showing Working Capital Requirements of

Current Assets	**Amount (₹)**
Stock of raw material (17,28,000 × 30/360)	1,44,000
Stock of work-in-progress [12,000 × (40 + 7.50 + 15)]	7,50,000
Stock of finished goods [24,000 × (40 + 15 + 30)]	20,40,000
Debtors for sale (6,12,000 × 60/360)	1,02,000
Cash	2,00,000
Total Current Assets (A)	**32,36,000**
Current Liabilities	
Creditors for purchase (18,72,000 × 30/360)	1,56,000
Creditors for wages (5,58,000 × 15/360)	23,250
Total Current Liabilities (B)	**1,79,250**
Net working capital (A – B)	**30,56,750**

Working Note-1

Statement of Cost

Particulars	**₹**
Opening stock of raw material	-
Add: Purchases (Bal. fig.)	18,72,000
Less: Closing stock of raw material (17,28,000 × 30/360)	(1,44,000)
Raw material consumed [(31,200 × 40) + (12,000 × 40)]	17,28,000
Add: Direct wages [(31,200 × 15) + (12,000 × 15 × 50%)]	5,58,000
Add: Overheads [(31,200 × 30) + (12,000 × 30 × 50%)]	11,16,000
Gross Factory Cost	34,02,000
Less: Closing work in progress [12,000 × (40 + 7.50 + 15)]	(7,50,000)
Cost of goods produced	26,52,000
Less: Closing stock of finished goods (26,52,000 × 24,000/31,200)	(20,40,000)
Cash cost of sales	6,12,000

8. Effect of Double Shift on Working Capital

- WIP stock will remain same

- Fixed cost will not change

- All elements i.e. RM stock, FG stock, Debtors, Prepaid expenses, outstanding exp. creditors will become double subject to price change if any.

- Units will get double due to which all dependent elements on units will get double.

Question – 7	

SK Ltd. has been operating its manufacturing facilities till 31.3.2022 on a single shift working with the following cost structure:

	Per unit (₹)
Cost of materials	6.00
Wages (out of which 40% fixed)	5.00
Overheads (out of which 80% fixed)	5.00
Profit	2.00
Selling price	18.00
Sales during 2021-22 - ₹ 4,32,000	

units = 24000

P.U.	Amt. (24000)	P.U.	Amt. (48000)
6	1,44,000	5.40	259200
3	72000	3	144000
2	48000	1	48000
4	96000	2	96000
1	24000	1	48000
16	384000	12.40	595200

As at 31.0.2022 the company held: Present

Stock of raw materials (at cost)	₹ 36,000
Work-in-progress (valued at prime cost)	₹ 22,000
Finished goods (valued at total cost)	₹ 72,000
Sundry debtors (at Sales)	₹ 1,08,000 $\times \frac{16}{18}$ = 96000

$\frac{36000}{6} \times 2 \times 5.4 = 54800$

$\frac{22000}{11} \times 9.40 = 18800$

$\frac{72000}{16} \times 12.40 \times 2 = 111600$

$\frac{96000}{16} \times 12.40 \times 2 = 148800$

In view of increased market demand, it is proposed to double production by working an extra shift. It is expected that a 10% discount will be available from suppliers of raw materials in view of increased volume of business. Selling price will remain the same. The credit period allowed to customers will remain unaltered. Credit availed from suppliers will continue to remain at the present level i.e. 2 months. Lag in payment of wages and expenses will continue to remain half a month.

You are required to prepare the additional working capital requirements, if the policy to increase output is implemented.

Solution

Present sales units = $\frac{17,28,000}{72}$ = 24,000 units

Sales units after double shift = 24,000 × 2 = 48,000 units

Statement of Cost

	24,000 units		48,000 units	
	Per unit	Total	Per unit	Total
Raw Material	6	1,44,000	5.40	2,59,200
Wages:				
Variable	3	72,000	3	1,44,000
Fixed	2	48,000	1	48,000
Overheads:				
Variable	1	24,000	1	48,000
Fixed	4	96,000	2	96,000
Total cost	16	3,84,000	12.40	5,95,200
Profit	2	48,000	5.60	2,68,800
Sales	18	4,32,000	18	8,64,000

Stock of raw material units on 31.3.2022 = $\frac{36,000}{6}$ = 6,000 units

Stock of WIP units on 31.3.2022 = $\frac{22,000}{11}$ = 2,000 units

Stock of finished goods units on 31.3.2022 = $\frac{72,000}{16}$ = 4,500 units

Debtors units on 31.3.2022 = $\frac{1,08,000}{18}$ = 6,000 units

Statement of Working Capital Requirement

	Single shift (24,000 units)			Double shift (48,000 units)		
	Units	Rate	Amount	Units	Rate	Amount
Raw Material stock	6,000	6	36,000	12,000	5.40	64,800
WIP stock	2,000	11	22,000	2,000	9.40	18,800
Finished goods stock	4,500	16	72,000	9,000	12.40	1,11,600
Sundry Debtors	6,000	16	96,000	12,000	12.40	1,48,800
Total Current Assets (A)			2,26,000			3,44,000
Creditors for material	4,000	6	24,000	8,000	5.40	43,200
Creditors for wages	1,000	5	5,000	2,000	4	8,000
Creditors for Overheads	1,000	5	450,000	2,000	3	6,000
Total Current Liabilities (B)			34,000			57,200
Working Capital (A – B)			1,92,000			2,86,800

Additional working capital requirement = ₹ 2,86,800 - ₹ 1,92,000 = ₹ 94,800.

9. Operating Cycle Method

$$\text{Amount of Working capital} = \frac{\textit{Annual operating cost}}{365} \times \textit{Operating Cycle Period}$$

$$\text{Number of operating cycles} = \frac{365}{\textit{Operating cycle period}}$$

10. Statement of Operating Cycle

Particulars	Days
Raw material period $\left[\frac{\textit{Average RM stock}}{\textit{RM Consumption per day}} \; or \; \frac{\textit{Average RM stock}}{\textit{Annual RM Consumption}} \times 365\right]$	
WIP Period $\left[\frac{\textit{Average WIP stock}}{\textit{Annual Cost of Production}} \times 365/52/12\right]$	
Finished goods period $\left[\frac{\textit{Average FG stock}}{\textit{Annual Cost of Goods Sold}} \times 365/52/12\right]$	
Receivables collection period $\left[\frac{\textit{Average Receivables}}{\textit{Annual Credit Sales}} \times 365/52/12\right]$	
Gross Operating Cycle	
(-) Creditors payment period $\left[\frac{\textit{Average Creditors}}{\textit{Annual Credit RM Purchases}} \times 365/52/12\right]$	
Net Operating Cycle	

Question – 8

Wint Limited is a mid-sized manufacturing company specializing in industrial equipment parts. It operates on a just-in-time procurement strategy but has recently faced delays in raw material deliveries due to supply chain disruptions in global markets. To maintain production continuity, the management has decided to maintain slightly higher inventories and build in a contingency reserve.

The following forecasted financial information is provided for the year ending 31st March, 2025:

Particulars	Balances as at 31st March, Cl. 2025 (₹in lakhs)	Balance as at 31st March, Op. 2024 (₹in lakhs)	Avg.
Raw Material	845	585	715
Work-in-progress	663	455	559
Finished goods	910	780	845
Receivables	1,755	1,456	1605.5
Payables	923	884	903.5

Additional details for the year ending 31st March, 2025:

- Annual purchases of raw materials (on credit): ₹5,200 lakh → Cr. Pd. = $\frac{903.5}{5200} \times 365 = 63$
- Annual cost of production: ₹5,850 lakh → WIP Pd. = $\frac{559}{5850} \times 365 =$

RM Cons. = 585 + 5200 − 845
= 4940

RM Pd. = $\frac{715}{4940} \times 365$

→ Fh PB. = $\frac{845}{6825} \times 365$

- Annual cost of goods sold (COGS): ₹6,825 lakh
- Annual operating cost (includes admin, manufacturing, distribution): ₹4,225 lakh → WC = $\frac{4225}{365} \times$
- Annual sales (on credit): ₹7,605 lakh → Drs. = $\frac{1605.5}{7605} \times 365$ =

(+) 10% NWC

The finance team is currently evaluating the company's working capital requirement due to a planned expansion and potential volatility in receivables collection caused by credit extensions to new clients.

Management wants to set aside a 10% contingency reserve in the working capital to deal with unforeseen cash flow constraints (e.g., delayed collections, inventory build-up, or supplier hold-ups). Assume one year = 365 days.

You are required to:

(i) Calculate the Net Operating Cycle Period (in days).

(ii) Determine the Number of Operating Cycles in the year.

(iii) Calculate the Working Capital Requirement including a 10% contingency reserve.

Solution

Working Notes:

1. **Raw Material Storage Period (R)**

$$\frac{\text{Average Stock of Raw Material}}{\text{Annual Consuption of Raw Material}} \times 365$$

$$\frac{\frac{585+845}{2}}{4,940} \times 365 = 52.83 \text{ or } 53 \text{ days}$$

Annual Consumption of Raw Material = Opening Stock + Purchases - Closing Stock

= ₹ 585 + ₹ 5,200 – ₹ 845 = ₹ 4,940 lakh

2. **Work – in - Progress (WIP) Conversion Period (W)**

$$= \frac{\text{Average Stock of WIP}}{\text{Annual Cost of Production}} \times 365$$

$$= \frac{\frac{455+663}{2}}{5,850} \times 365 = 34.87 \text{ or } 35 \text{ days}$$

3. **Finished Stock Storage Period (F)**

$$= \frac{\text{Average Stock of Finished Goods}}{\text{Cost of Goods Sold}} \times 365$$

$$= \frac{\frac{780+910}{2}}{6,825} \times 365 = 45.19 \text{ or } 45 \text{ days.}$$

4. **Receivables (Debtors) Collection Period (D)**

$$= \frac{\text{Average Receivables}}{\text{Annual Credit Sales}} \times 365$$

$$= \frac{\frac{1,456 + 1,755}{2}}{7,605} \times 365 = 77.06 \text{ or } 77 \text{ days}$$

5. **Payables (Creditors) Payment Period (C)**

$$= \frac{\text{Average Payables for Materials}}{\text{Annual Credit Purchases}} \times 365$$

$$= \frac{\frac{884 + 923}{2}}{5,200} \times 365 = 63.42 \text{ or } 63 \text{ days}$$

(i) Net Operating Cycle Period

= R + W + F + D - C

= 53 + 35 + 45 + 77 – 63 = 147 days

(ii) Number of Operating Cycles in the Year

$$= \frac{365}{\text{Operating Cycle Period}} = \frac{365}{147} = 2.48 \text{ times}$$

(iii) Amount of Working Capital Required

$$= \frac{\text{Annual Operating Cost}}{\text{Number of Operating Cycles}} = \frac{4,225}{2.48} = ₹\ 1,703.63 \text{ lakh}$$

Add: 10% Contingency Reserve = ₹ 170.363 lakh

Total Working Capital Requirement (including contingency) = ₹ 1,874 lakh

Question – 9	

The following information is provided by MNP Ltd. for the year ending 31st March, 2020:

Raw Material Storage Period	+ 45 days
Work-in-Progress conversion period	+ 20 days
Finished Goods storage period	+ 25 days
Debt Collection period	+ 30 days
Creditors' payment period	– 60 days
Annual Operating Cost	₹ 25,00,000
(Including Depreciation of ₹ 2,50,000)	

→ op. cycle = 60 days

Assume 360 days in a year.

You are required to calculate:

(i) Operating Cycle period → 60 days

(ii) Number of Operating Cycle in a year → $\frac{360}{60} = 6$ cycles

New op. cycle = 60 - 30 = 30 days

New WC = $\frac{(25L - 2.50L) \times 30}{360}$

= 187500

Red. in WC = 3.75L – 187500

= 187500

$\frac{(25L - 2.50L)}{360} \times 60 = 3.75L$

(iii) Amount of working capital required for the company on a cost basis.

(iv) The company is a market leader in its product and it has no competitor in the market. Based on a market survey it is planning to discontinue sales on credit and deliver products based on pre-payments in order to reduce its working capital requirement substantially You are required to compute the reduction in working capital requirement in such a scenario.

Solution

(i) **Statement showing Operating cycle**

	Raw Material storage Period	= 45 days
	WIP Conversion Period	= 20 days
	Finished goods storage period	= 25 days
	Debt collection period	= 30 days
Less:	Creditors' payment period	= (60 days)
	Operating cycle period	= 60 days

(ii) Number of operating cycles in a year $= \frac{360}{Operating\ cycle\ period} = \frac{360}{60\ days} = 6$ cycles

(iii) Amount of working capital required on cash cost basis $= \frac{(25,00,000-2,50,000)}{6} =$ ₹ 3,75,000

(iv) New operating cycle period = 60 days – Debt collection period = 60 – 30 = 30 days

Number of operating cycles in a year $= \frac{360}{30} = 12$ cycles

New amount of working capital required on cash cost basis

$= \frac{(25,00,000-2,50,000)}{12} =$ ₹ 1,87,500

Saving in cash cost of working capital = ₹ 3,75,000 - ₹ 1,87,500 = ₹ 1,87,500

11. Maximum Permissible Bank Finance (MPBF)

Norm – 1) MPBF = 75%(CA – CL)

Norm – 2) MPBF = (75% × CA) – CL

Norm – 3) MPBF = [75% × (CA – Hard core CA)] – CL

Norm-3) MPBF = (75%) (480 - 30) – 180 = 157.50

Note – Existing bank finance if any should be excluded from CL.

Norm-1) MPBF = 75%(480 - 180) = 225

Norm-2) MPBF = (75%) (480) – 180 = 180

Question – 10	

Balance sheet of X Ltd. for the year ended 31st March, 2022 is given below:

(₹ in lakhs)

Liabilities	Amount	Assets	Amount
Equity Shares ₹ 10 each	200	Fixed Assets	500
Retained Earnings	200	Raw materials	150
11% Debentures	300	WIP	100
Public Deposits (Long Term)	100	Finished goods	50
Trade Creditors	80	Debtors	125
Bills Payable	100	Cash/Bank	55
	980		980

CL =180 CA =480

Calculate the amount of maximum permissible bank finance under three methods as per Tandon Committee lending norms. The total core current assets are assumed to be ₹ 30 lakhs.

Solution

Total current assets = 150 + 100 + 50 + 125 + 55 = ₹ 480 lakhs

Total current liabilities = 80 + 100 = ₹ 180 lakhs

Core current assets = ₹ 30 lakhs

1st Method

		(₹ in lakhs)
	Total current assets required	480
Less:	Current liabilities	(180)
	Working capital gap	300
Less:	25% of long term sources	(25)
	Maximum permissible bank borrowings	**225**

2nd Method

	Current assets required	480
Less:	25% to be provided by long term funds	(120)
		360
Less:	Current Liabilities	(180)
	Maximum permissible bank borrowings	**180**

3rd Method

	Current assets	480
Less:	Core current assets required	(30)
		450
Less:	25% to be provided by long term funds	(112.50)
		337.50
Less:	Current Liabilities	(180)
	Maximum permissible bank borrowings	**157.50**